(Y-Shield): High-Level Technical Architecture

Yochan-Shield is a **Sovereign Post-Quantum Cryptography (PQC) Gateway** designed to provide 2026-standard security against "Harvest Now, Decrypt Later" (HNDL) attacks.

---

1. Core Architecture: The "Hybrid" Proxy

The system operates as a **PQC-Native Reverse Proxy** that sits in front of existing infrastructure (Legacy APIs, Web Servers).

- **Gateway (Data Plane)**: Built on a custom-compiled **Nginx 1.26.2** linked with **OpenSSL 3.4.0**.
- **Cryptographic Engine**: Integrates the **Open-Quantum-Safe (OQS) Provider**, enabling NIST-standardized lattice-based mathematics.
- **Dual-Signature Stack**: The gateway simultaneously serves **ML-DSA-65** (Quantum-Safe) and **RSA-4096** (Classical) certificates. This ensures 100% backward compatibility with legacy browsers while providing future-proof security for modern clients.

---

2. Cryptographic Standards (2026 Sovereign)

Y-Shield implements the latest **NIST FIPS** and **CNSA 2.0** requirements:

- **Key Encapsulation (KEM)**: x25519_mlkem768 (Hybrid). Combines Elliptic Curve Diffie-Hellman with ML-KEM (Kyber) for guaranteed security even if one algorithm is compromised.
- **Digital Signatures**:
    - **ML-DSA (Dilithium)**: Used for high-speed session identity and authentication.
    - **SLH-DSA (SPHINCS+)**: Used for long-term document integrity (stateless signing) for assets meant to last 20+ years.
- **Groups**: Support for x25519_mlkem768, p256_mlkem768, and falcon512.

---

3. The Management Plane (Dashboard & AI)

A **Next.js 15** control center that manages the security posture of the network.

- **Quantum Threat Scanner**: A real-time engine that performs deep network probes to identify "Harvest-Ready" (vulnerable) endpoints.
- **CBOM Generator**: Automatically produces a **Cryptographic Bill of Materials**, a mandatory 2026 requirement for federal and sovereign compliance.
- **Sovereign Kill Switch**: An administrative fail-safe that can deprecate compromised PQC algorithms across the entire gateway cluster without downtime.
- **Audit Ledger**: A **PostgreSQL/Prisma** backed immutable log of all PQC handshakes and cryptographic events.

---

4. Compliance & Sovereignty

- **Indian NQM Level 4**: Configured to meet the **National Quantum Mission's** highest assurance level, including "Sovereign Node Locking" which ensures KEM operations remain within domestic jurisdiction.
- **NIST 800-207**: Implements Zero-Trust principles through continuous identity verification via **Email OTP** and **Mobile-Bound Licensing**.

---

5. Deployment Stack

- **Containerization**: Multi-stage Docker builds to keep the production image lightweight while including the full OQS/OpenSSL toolchain.
- **Orchestration**: Docker Compose for local/edge deployment; **Terraform/AWS** ready for cloud-scale infrastructure.
- **Backend**: Next.js API Routes (Serverless ready) with **Supabase/PostgreSQL** for persistence.

In essence, Y-Shield is a plug-and-play security layer that upgrades any legacy web application to "Quantum-Immune" status in minutes.